

Product Mapping Schema Documentation

Malineni Vagdevi

Nainisha Bandari

Branch:

Feature/product-mapping

Objective:

The Product Mapping schema is designed to support AI-powered product compatibility checks, avatar-product matching, recommendation systems, and virtual try-on functionality.

File Created:

Models/ProductMapping.js

Schema Fields:

Field	Type	Description
productId	ObjectId	References the associated product
category	String	Product category (Men, Women, Kids, etc.)
clothType	String	Type of clothing (Shirt, T-Shirt, Dress, Jeans, etc.)
supportedBodyTypes	Array[String]	Body types supported by the product
compatibility	String	Compatibility status for AI processing
avatarCompatibility	Array[String]	Supported avatar types
createdAt	Date	Record creation timestamp

Compatibility Workflow

User Avatar



Backend Checks Compatibility



Matching Products Selected



AI Try-On Starts

Workflow Explanation

1. User uploads or selects an avatar.
 2. Backend retrieves avatar attributes.
 3. ProductMapping collection is queried.
 4. Compatible products are filtered based on:
 - category
 - cloth type
 - body type
 - avatar compatibility
 5. Matching products are returned.
 6. AI Try-On service generates the preview.
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AI Integration Purpose

The Product Mapping schema provides structured compatibility data required by AI services.

Features Supported

- Avatar-to-product matching
- AI virtual try-on validation
- Product filtering
- Compatibility scoring
- Personalized recommendations

Benefits

- Improved try-on accuracy
 - Faster compatibility checks
 - Better recommendation quality
 - Reduced AI processing errors
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Recommendation Engine Usage

The schema can be used by recommendation services to:

- Suggest products based on body type

- Recommend compatible clothing categories
- Filter unsupported products
- Personalize shopping experiences

Example Use Case

If a user has an athletic body type and uses a male avatar, the system can retrieve products that support:

supportedBodyTypes = Athletic

avatarCompatibility = Male Avatar

and recommend only compatible items.

Testing Performed

Backend Startup Test

Command: Npm run dev

Expected Result:

MongoDB Connected

Server Running Successfully

Validation Test

Verified:

- ProductMapping schema loads successfully

- No schema compilation errors
 - No startup failures
 - MongoDB connection remains functional
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Deliverables

- ProductMapping.js created
- Schema fields implemented
- Compatibility workflow documented
- AI integration purpose documented
- Recommendation engine support documented
- Backend tested successfully